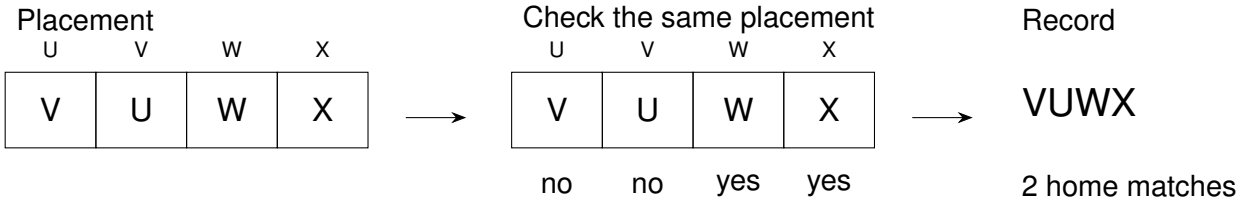
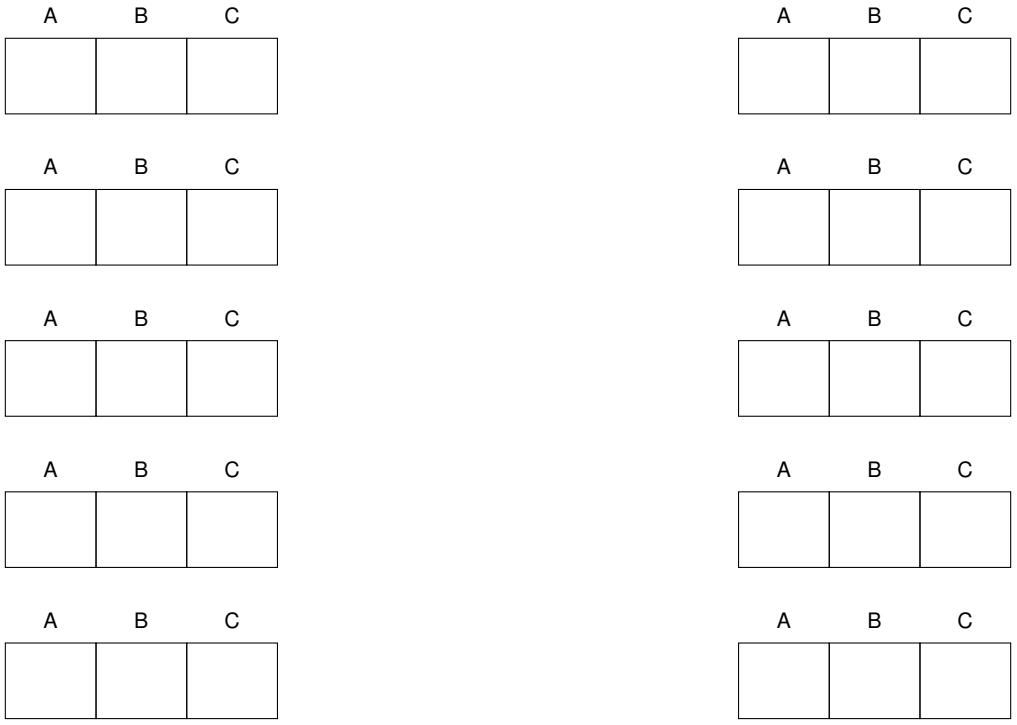
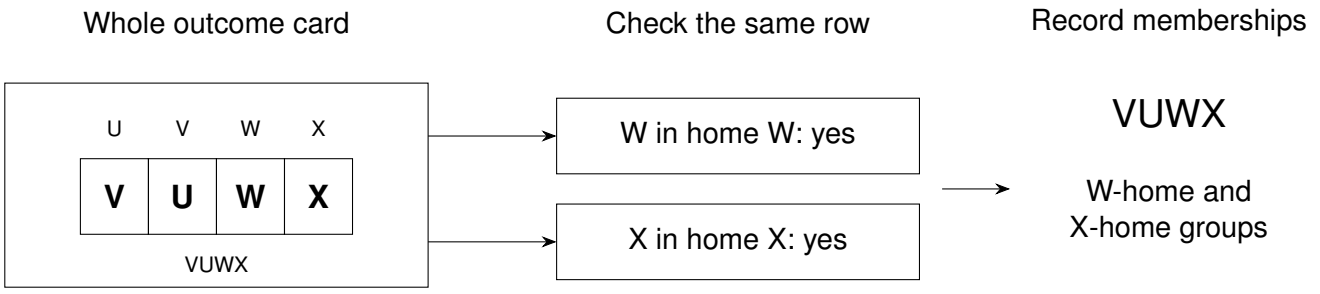
Use a card set and the home strip with the same letters. Use each card once, with one card in every home. Keep the home labels fixed. A *home match* means the card and its home have the same letter. Check a complete row before moving any card. Record the cards from left to right; different rows count separately. A row with no home matches is *home-free*.



Problem 1: Find every row of A, B, C with no home matches. Can a row of these three cards have exactly two home matches?



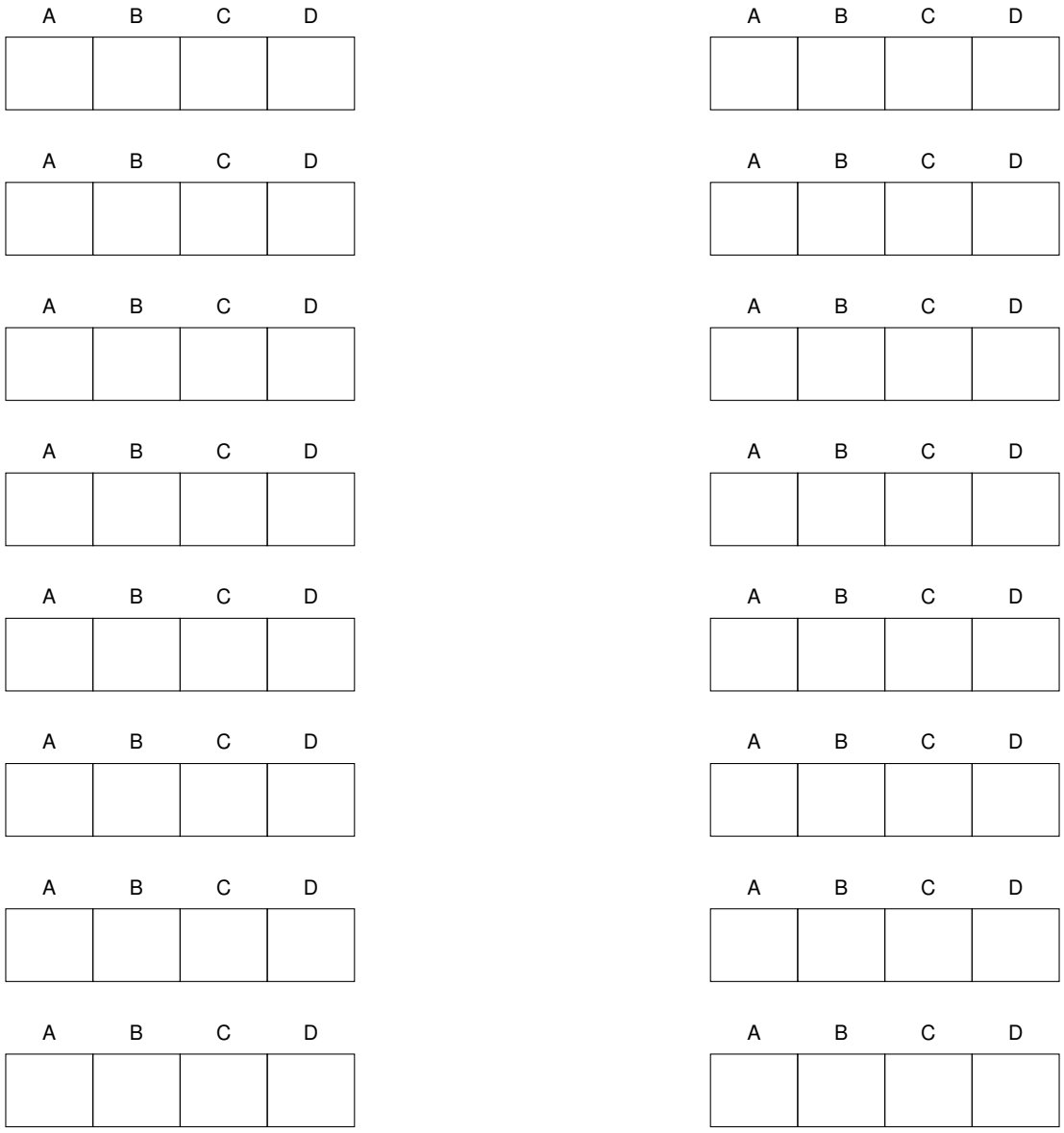
One outcome card is one whole placement. It can belong to several groups at once. Group the outcome cards on the table.



Problem 2: Use the six A–C outcome cards. Make one group with A in home A and another with B in home B. A card may belong to both groups. How many rows are outside both groups? C may stay home.

A student counts $6 - 2 - 2 = 2$. Repair this count and show what went wrong.

Problem 3: Find every A–D row with no home matches. Make a catalog your table can check for repeated or missing rows.



Problem 4: Count the A–D rows with A away from home A and B away from home B. C and D may stay home. Use the 24 outcome cards to check the count $24 - 6 - 6$. How should the count be repaired?

Problem 5: There are four home-groups: A at A, B at B, C at C, and D at D. Each contains six of the 24 A–D rows. A student tries $24 - 6 - 6 - 6 - 6$ to count the rows outside every group. Repair the count using overlaps.

An overlap includes every row with the required matches, even when other cards are also home.